

15. NEURAL CRESTS

DURATION : 03:29 STUDY SHEET

COURSE SUMMARY

This video focuses on neural crest movement, a key process in embryology that gives rise to various bodily structures after neurulation. It explores types of movement, including prosencephalic, mesencephalic, and rhombencephalic, and explains how the neural crest challenges our understanding of genetic and metabolic interactions during embryonic development. By learning to identify signs in body structures, particularly in the face, therapists can better interpret patients' internal states. The video also emphasises the importance of cell migrations and environmental influences in tissue formation, highlighting the role of micro-vessels and micro-nerves, to provide an integrated view of biodynamic embryology.

NEURAL CREST MOVEMENT

The **neural crest movement** is a complex process involving different types of **differentiation** and **induction factors**. These factors can be **metabolic** or **passive** in nature, and intervene after **neurulation**.

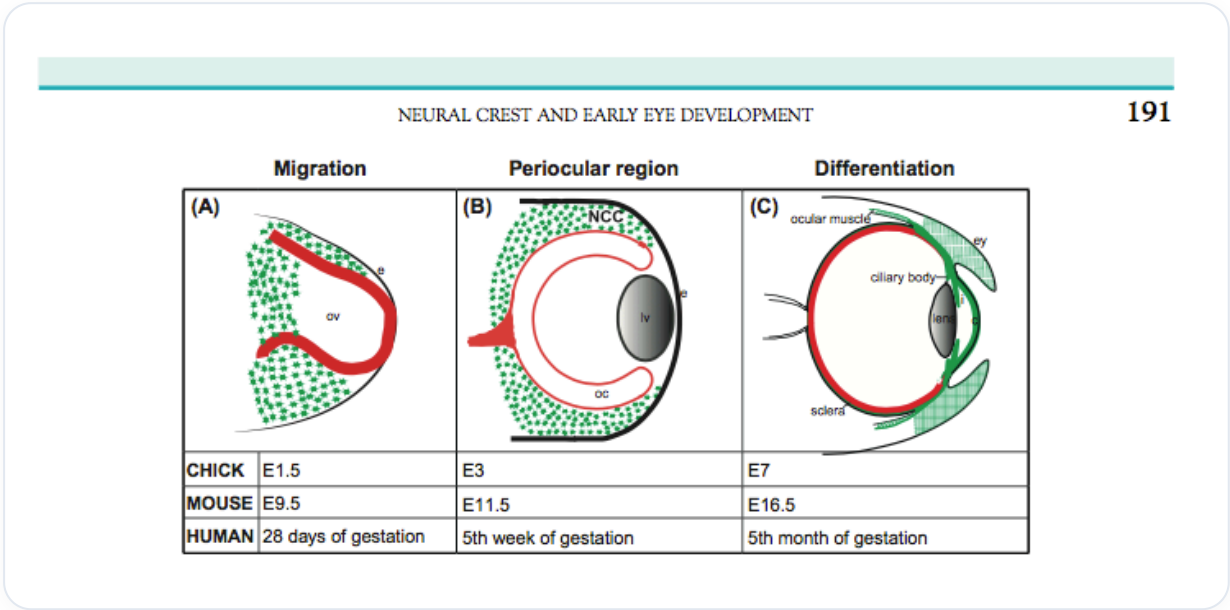


FIG. — Neural crest movement

Blood vessels play a crucial role in guiding the system, while **genetic** elements also influence the orientation of cells towards the appropriate areas.

TYPES OF MOVEMENTS

There are three main types of movements:

- **Prosencephalic**
- **Mesencephalic**
- **Rhombencephalic**

During the **closure of the neural tube**, the cells that will form the neural crest emerge. These cells will give rise to various processes in the body, creating a **flow** that extends from back to front, invading the entire **face**. This means that the face, bones, skin, down to the level of the **platysma**, originate from the neural crest.

This tissue is highly **informed** and **informative**. For example, everything in your abdomen also originates from the neural crest. **Ganglia** and other structures reflect the internal state of the body, allowing one to learn to **read the face** through wrinkles, dark circles, and colours.

MIGRATION AND DEVELOPMENT

The **prosencephalon** develops into the **telencephalon** and **diencephalon**, with significant migration of the diencephalon. The latter, as well as the **diencephalon** and the anterior **mesencephalon**, migrate towards the neural crest. This flow process is essential for development.

Before the closure of the neural tube, negative influences, such as those related to the **hepatopancreatic, thyroid, and cardiac systems**, provide information to the neural crest, which is also considered **ectomesenchyme**. The latter is a mixture of **mesoderm** and neural crest.

IMPORTANCE OF THE NEURAL CREST

The neural crest plays a fundamental role in **migration** and **colonisation**, contributing to specific characteristics. It also influences the construction of **micro-vessels** and **micro-nerves**, particularly **pericytes**.

This process allows for the creation of support and emergence points, providing an **ambient embryonic force** that restructures growing micro-nerves. It is important to note that these interactions occur at an **electrical** level.

At the level of the eye, a green area encompasses and shapes the anterior tissue, which is **ectoderm**, in relation to this mesodermal flow, particularly around the **sclera**.

